

SpectraLab: spectral measurement, analysis and provenance

A concise technical presentation of a base-MATLAB package for measuring, preserving, analysing and communicating spectral data

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Purpose. SpectraLab is a system for scientific spectral measurement, not merely a program for drawing curves. It connects the physical instrument, measurement, durable archive, defined analyses and traceable reports. Its governing principle is: *Measure once. Save forever. Verify always.*



Figure 1. SpectraLab separates measurement, preservation, analysis and presentation.

1. What SpectraLab does

SpectraLab receives calibrated spectral values from a connected spectrophotometer system and turns them into a long-lived scientific record. A spectrum consists of a strictly increasing wavelength vector in nanometres and one finite measured value for each wavelength. Its label, unit, timestamp and metadata state what the measurement means.

Measurement and instrument

The verified workflow uses an X-Rite i1Pro2 or an original GretagMacbeth Eye-One Pro Rev. B through the ArgyllCMS program *spotread*. Python, pexpect and ptyprocess control the bounded external process. SpectraLab waits for the correct instrument state, tells the operator when to place the white reference or sample, and accepts a spectrum only after the data and process outcome have been validated.

Important boundary. The instrument and ArgyllCMS are authoritative for calibration status and physical measurement. SpectraLab cannot replace clean optics, correct geometry, sufficient warm-up, correct placement or instrument accuracy. Incomplete measurements and calibration-only output never become scientific measurement archives.

Responsibilities remain separate

Layer	Primary responsibility
Instrument	Detector, optics, internal calibration and physical response
ArgyllCMS	Instrument communication, measurement mode and returned spectrum
SpectraLab	Workflow, validation, archive, analysis, report and provenance
Operator	Geometry, placement, warm-up, sample identity and scientific judgement

This separation prevents a plotting routine from creating new scientific meaning and prevents an example script from introducing a competing calculation. A derived product never replaces the measurement that supports it.

2. From measurement to durable result

Immutable archives and identity

An accepted measurement is saved as a stable MAT archive. The archive contains wavelengths, values, units, instrument and calibration information, measurement mode, quality evidence and metadata. A UUID identifies the archive instance. A deterministic SHA-256 hash identifies the scientific content and changes if measured values or other normative information change.

Identity	Meaning
Archive UUID	One particular archive instance
ContentHash	Deterministic identity of the scientific content
Metadata	Operator, project, sample, comment and declared context
Processing history	Provenance for later calculations; never a replacement for source values

The native archive is authoritative. JSON, CSV and text are readable or tabular exchange formats, but are not automatically as complete as the original archive. Comments and filenames may change without changing the scientific content hash.

Wavelengths, interpolation and integration

Reference/sample comparisons require a common wavelength domain. Identical wavelength grids are required by default. If grids differ, the user must explicitly permit resampling. SpectraLab uses only the common measured interval and performs no silent extrapolation. The default method is shape-preserving cubic interpolation, pchip, with refinement factor four.

Interpolated points are not new observations. They are a numerical aid for comparison and integration within the measured domain. Reports therefore state whether resampling occurred, the method, refinement and effective wavelength range.

Registered analyses

A canonical analysis registry is the authority for reportable analyses. Each entry declares its analysis ID, inputs, method, standard, result fields and figure contract. Unknown or incomplete definitions fail before report generation.

Analysis group	Examples of results
Spectral comparison	Transmission, optical density, mean and signed difference
Densitometry	White, ISO visual, Status A and Status M RGB + W
Colorimetry	CIE XYZ, xyY, CIELAB and ColorChecker summary
Light sources	Spectrum, peak, integral, CCT, Duv and CIE 13.3 CRI

Reports and presentation

Interactive figures and PNG files may show rapid measurement information such as peak, integral, project, operator and instrument. In PDF reports, authoritative numerical results and provenance belong in text and tables, while figures contain the plot, title and legend. The MAT/JSON archive remains the scientific source.

Presentation must never define the calculation. A polished figure is not scientific proof by itself; values must come from registered result fields and source hashes must remain valid.

3. Traceable use and limitations

ColorChecker as spectral reference evidence

A ColorChecker session consists of a JSON manifest and one immutable reflectance archive for each patch. Target model, geometry, patch count, orientation and order can be locked. An incorrect patch is not overwritten; controlled remeasurement creates a new archive and amendment manifest while retaining the original record.

Measured reflectance $R(\lambda)$ is primary. XYZ and Lab are derived values and do not replace the spectrum. Camera-41 or another project can therefore recalculate colour under another declared illuminant and observer without remeasuring the target.

Provenance and reproducibility

SpectraLab records instrument name and serial number, calibration status, operator, timestamp, measurement kind, resolution, backend, wavelengths, values, units and quality. Reports add source filenames and content hashes. Derived products record analysis identity, definition version, input order, interpolation and wavelength range.

Risk	Control
File renamed	ContentHash is tied to the payload, not the filename
Archive modified	Validation recomputes and compares SHA-256
Analysis changed	Analysis ID, definition version and source hash identify the calculation
Different wavelength grids	Default failure or explicitly recorded resampling

Reproducibility means that the same archived scientific content and declared processing produce the same numerical result within stated tolerance. It does not mean that a new physical measurement must be identical; instrument repeatability, positioning, sample variation, source stability and environment remain experimental variables.

Base MATLAB and external dependencies

Standard/base MATLAB is the only permitted MathWorks product. Report Generator and other toolboxes are forbidden dependencies. ArgylCMS, Python, pexpect and ptyprocess are permitted external components with clearly bounded and versioned roles. Their resolved paths, versions and command options belong to measurement provenance.

Overall conclusion

Principle	Consequence
MEASURE ONCE	The original spectrum is preserved and can be analysed again.
SAVE FOREVER	Stable archives and content hashes keep results independent of presentation.
VERIFY ALWAYS	Calibration, format, analysis contract, source identity and report are checked.

Conclusion. SpectraLab provides an auditable chain from a calibrated instrument result to a durable spectrum, explicit analysis and readable report. It replaces neither instrument metrology nor operator judgement; it preserves information, declares every transformation and refuses to create trustworthy-looking results from incomplete evidence.

Sources

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[6] The MathWorks, MATLAB; Python Software Foundation, Python.